Rodent training data
4,539 profiles
96 datasets

Modules
23, defined by co-expression

Clocks
module and composite, as released

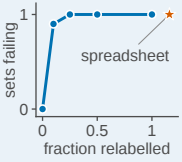
Human lung scRNA-seq
IPF, COPD and control; two cohorts

Ortholog mapping, pseudobulks
donor × cell type

Adopted from previous work
Re-implemented
New in this study
✓ Tested against known truth

1 Input check

Are the module gene labels correct?

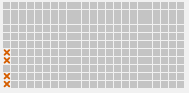


Serialized clocks pass; the published spreadsheet labels fail. With 10% of genes relabelled, 18 of 20 sets fail.

✓ memberships with relabelled genes

2 Score

Every clock scored in donor × cell-type pseudobulks



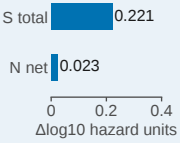
253 planned tests; 4 (crosses) are not estimable, leaving 249.

11 cell types × 23 modules

GSE136831: 32 IPF, 28 control, 18 COPD donors

3 Decompose

Does an unchanged composite hide opposing module signals?



In IPF the composite barely moves (N = 0.02) while its components do (S = 0.22; 13 of 24 at FDR < 0.10).

permutation null and donor bootstrap

4 Claim gates

Is a cell-type × module effect robust, specific to IPF and to the module?

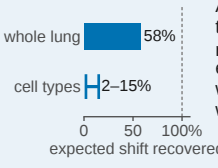
249 tests
10 hits
9 robust
2 IPF-specific
0 passed

Robust inference, a COPD contrast and competitive nulls applied across all tests. No IPF effect passed all gates.

✓ spike-ins: 0 of 19,334 null pairs passed

5 Resolution

Does the clock carry age signal at the resolution it is read?

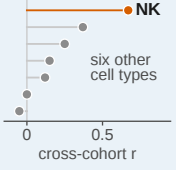


Aging mouse lung: the composite recovers 58% of the expected shift in whole lung, 2–15% within cell types.

✓ 3- versus 24-month mice

6 Replication

Does the effect pattern hold in a second cohort?



Only the NK effect pattern agrees across cohorts (r = 0.67, p = 0.008); no individual effect replicates.

✓ 300 simulated cohort pairs per scenario